

**ABSTRACT**

This application report contains material applicable to the LPDDR4 interface of Jacinto7 AM6x/TDA4x/DRA8x processor board designs.

Table of Contents

1 Overview	3
1.1 Supporting Documentation	3
1.2 Board Designs Supported	3
1.3 General Board Layout Guidelines	4
1.4 PCB Stack-Up	5
1.5 Bypass Capacitors	6
1.6 Velocity Compensation	7
2 LPDDR4 Board Design and Layout Guidance	8
2.1 LPDDR4 Introduction	8
2.2 LPDDR4 Device Implementations Supported	8
2.3 LPDDR4 Interface Schematics	8
2.4 Compatible JEDEC LPDDR4 Devices	12
2.5 Placement	12
2.6 LPDDR4 Keepout Region	13
2.7 Net Classes	13
2.8 LPDDR4 Signal Termination	13
2.9 LPDDR4 VREF Routing	14
2.10 LPDDR4 VTT	14
2.11 CK, CMD_ADDR, and CTRL Topologies	14
2.12 Data Group Topologies	15
2.13 CK, CMD_ADDR, and CTRL Routing Specification	16
2.14 Data Group Routing Specification	18
2.15 Channel, Byte, and Bit Swapping	19
3 LPDDR4 Board Design Simulations	19
3.1 Board Model Extraction	19
3.2 Board-Model Validation	19
3.3 S-Parameter Inspection	19
3.4 Time Domain Reflectometry (TDR) Analysis	20
3.5 Simulation Integrity Analysis	20
3.6 Design Example	26
4 Revision History	30

List of Figures

Figure 2-1. 32-Bit, Single-Rank LPDDR4 Implementation	9
Figure 2-2. 32-Bit, Dual Rank LPDDR4 Implementation	10
Figure 2-3. 16-Bit, Single Rank LPDDR4 Implementation	11
Figure 2-4. TDA4VM/DRA829 LPDDR4 Example Placement (Top View)	12
Figure 2-5. Example LPDDR4 Keepout Region for TDA4VM/DRA829 (Top View)	13
Figure 2-6. LPDDR4 CK Topology	14
Figure 2-7. LPDDR4 CMD_ADDR, CTRL Topology/Branched Signals	14
Figure 2-8. LPDDR4 CTRL Topology/Point-to-Point Signals	14
Figure 2-9. LPDDR4 DQS Topology	15
Figure 2-10. LPDDR4 DQ/DM Topology	15
Figure 3-1. TDR Plot Example With Impedance Mismatch	20

Figure 3-2. Typical System-Level DDR Schematic.....	21
Figure 3-3. Example of Ring-back Margin Waveforms (High/Low).....	23
Figure 3-4. Vix_CK and Vix_DQS Ratio for Eye Quality.....	23
Figure 3-5. Sample Simulated LPDDR4-4266 Read Eye With Diamond-Shaped Eye Mask.....	25
Figure 3-6. Sample Simulated LPDDR4-4266 Write Eye With Rectangular JEDEC Eye Mask.....	25
Figure 3-7. Sample Simulated LPDDR4-4266 CA Eye With Rectangular JEDEC Eye Mask.....	26
Figure 3-8. Example LPDDR4 Clock and CA Routing.....	27
Figure 3-9. Example LPDDR4 Data Byte(s) and DQS(s) Routing.....	27
Figure 3-10. Example LPDDR4 Trace Impedance Scan.....	28
Figure 3-11. LPDDR4 Simulation Results for CA.....	29
Figure 3-12. LPDDR4 Simulation Results for Write.....	29
Figure 3-13. LPDDR4 Simulation Results for Read.....	30

List of Tables

Table 1-1. PCB Stack-up Specifications	5
Table 1-2. Bulk Bypass Capacitors.....	6
Table 1-3. High-Speed Bypass Capacitors.....	7
Table 2-1. Supported LPDDR4 SDRAM Combinations.....	8
Table 2-2. Compatible JEDEC LPDDR4 Devices.....	12
Table 2-3. LPDDR4 Placement Recommendations.....	12
Table 2-4. Clock Net Class Definitions.....	13
Table 2-5. Signal Net Class Definitions.....	13
Table 2-6. CK, CMD_ADDR, and CTRL Routing Specifications.....	16
Table 2-7. Data Group Routing Specifications.....	18
Table 3-1. Example Data Write ODI/ODT Optimization.....	22
Table 3-2. Vix_CK and Vix_DQS Ratio for Eye Quality.....	24
Table 3-3. LPDDR4/4x Eye Mask Definitions/Requirements.....	24
Table 3-4. Example PCB Stackup for LPDDR4.....	26
Table 3-5. LPDDR4 Performance Impact From StackUp.....	27
Table 3-6. LPDDR4-4266 Layer Impact Performance (Read at Pad)	28
Table 3-7. LPDDR4-4266 Layer Impact Performance (Write at Pad).....	28
Table 3-8. Example LPDDR4 Trace Impedance Summary for Data.....	28
Table 3-9. Example LPDDR4 Trace Impedance Summary for CA.....	28
Table 3-10. Example LPDDR4 Simulation Results From Improved Trace Impedance.....	29

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1 Overview

The Jacinto7 AM6x/TDA4x/DRA8x family of processors support LPDDR4 memory interfaces. This document contains specific information on how to design and implement LPDDR4 memory interfaces with this family of devices.

To facilitate software configuration of the LPDDR4 subsystem, please refer the Jacinto7 DDRSS Register Configuration Tool ([Link](#)).

1.1 Supporting Documentation

The Jacinto7 AM6x/TDA4x/DRA8x LPDDR4 Design Guidelines application note supports the following devices:

- J7200
 - DRA821
- J721E
 - TDA4VM
 - DRA829
- J721S2
 - AM68x
 - TDA4VE
 - TDA4AL
 - TDA4VL
- J722S
 - AM67x
 - TDA4VEN
 - TDA4AEN
- J742S2
 - TDA4VPE
 - TDA4APE
- J784S4
 - AM69x ✓
 - TDA4VH
 - TDA4AH
 - TDA4VP
 - TDA4AP

For AM6x parts not listed (for example AM62P), refer to the application note provided on those product pages, for example: [AM62Ax/AM62Px LPDDR4 Board Design and Layout Guidelines](#)

1.2 Board Designs Supported

In order to achieve the high frequency targets for Jacinto7 AM6x/TDA4x/DRA8x LPDDR4 interfaces, an optimal PCB implementation is required. TI highly recommends that customer design copies the LPDDR4 PCB of the specific device EVM exactly, and in every detail (PCB material, routing, spacing, vias w/ back-drill, and so forth) in order to achieve the full specified interface frequency/data rate. If the design does not or cannot copy the TI solution, TI's EVM should still be used as a starting point/reference. Based on any compromises made, the customer design may need to constrain the interface frequency/data rate.

The goal of this document is to define a set of layout, routing, and simulation rules that allow designers to successfully implement a robust design for the topologies that TI supports. It is also required that the PCB design be simulated to ensure the design targets are achieved. TI will limit debug/support for boards that have not been designed and simulated according to the steps defined in this document. Systems that do not follow the TI EVM implementations and/or do not have valid simulation results will likely need to run at a reduced DDR frequency.

This document provides Reference eye masks as guidance for validation of the simulations results. It is still expected that the PCB design work (design, layout, and fabrication) is performed and reviewed by a highly knowledgeable high-speed PCB designer. Problems such as impedance discontinuities when signals cross a split in a reference plane can be detected visually by those with the proper experience.

TI only supports board designs that follow the guidelines in this document. These guidelines are based on well-known transmission line properties for copper traces routed over a solid reference plane. Declaring insufficient PCB space does not allow routing guidelines to be discounted. TI will limit debug/support for designs that have not been simulated according to the steps defined in this document.

1.3 General Board Layout Guidelines

To ensure good signaling performance, the following general board design guidelines must be followed:

- Always follow TI's example layouts/EVM designs as close as possible. If concepts or routing strategies are not understood, questions can be posted on E2E.
- All signals need ground reference (strongly suggest on both sides). Maintain the common ground/reference for all signals and for all bypass/decoupling capacitors.
- Avoid crossing plane splits in the signal reference planes.
- Use the widest trace that is practical between decoupling capacitors and memory modules.
- Minimize inter-symbol interference (ISI) by keeping impedances matched. This is especially true for the 'T-branch' signals where trace widths are adjusted to match trace impedance.
- Minimize crosstalk by isolating sensitive signals, such as strobes and clocks, and by using a proper PCB stack-up. Add additional spacing for clock and strobe traces to also help minimize crosstalk.
- Avoid return path discontinuities by adding vias or capacitors whenever signals change layers and reference planes.
- Minimize reference voltage noise through proper isolation and proper use of decoupling capacitors on the reference input pins on the SDRAMs.
- Keep the signal routing stub lengths as short as possible.
- Route all signals as strip-line. Avoid micro-strip traces except for BGA break-out areas.
- Via-to-via coupling can be significant part of PCB-level crosstalk. Dimension and pitch of vias is important. For high speed interfaces, consider GND shielding vias. This via coupling is one factor for recommending data signals be routed on layers closest to processor.
- Via stubs affect signal integrity. Via back-drilling can improve signal integrity, and may be required in some instances.

For more information, see the [High-Speed Interface Layout Guidelines Application Report](#). It provides additional general guidance for successful routing of high-speed signals.

1.4 PCB Stack-Up

The typical stack-up for routing the DDR interface is a ten layers. However, this can only be accomplished on a board with routing room with large keep-out areas. Additional layers are required if:

- The PCB layout area for the DDR Interface is restricted, which limits the area available to spread out the signals to minimize crosstalk.
- Other circuitry must exist in the same area, but on layers isolated from the DDR routing.
- Additional planes layers are needed to enhance the power supply routing or to improve EMI shielding.

Board designs that are relatively dense may require more layers to properly allow the DDR routing to be implemented such that all rules are met.

All DDR signals must be routed adjacent to a solid VSS reference plane. When multiple VSS reference planes exist in the DDR routing area, stitching vias must be implemented nearby wherever vias transfer signals to a different VSS reference plane. This is required to maintain a low-inductance return current path.

It is strongly recommended all DDR signals be routed as strip-line. Some PCB stack-ups implement signal routing on 2 adjacent layers. This is not recommended as crosstalk occurs on any trace routed parallel to another trace on an adjacent layer, even for a very short distance. It is recommended to route LPDDR4 signals on PCB layers closer to the SoC within the stackup, giving the signal a shorter travel time through the via. The PCB layers farther from the SoC will have longer travel times through the via, which can increase coupling between vias. Both signal and via coupling can lead to smaller timing margins.

Note a shorter via travel could mean a longer via stub (if using standard drill vias), so that is to be considered as well. Simulation can be used to determine if via stub length is an issue.

PCB material is another important factor. Depending on the design specifics, it may be required to use a higher frequency material such as ISOLA I-Speed or equivalent/better to achieve highest data rates (4266 Mbps). Standard FR4 products like 370HR can be used for lower data rates. In specific cases, it is sufficient for higher data rates as well.

Table 1-1. PCB Stack-up Specifications

Number	Parameter ⁽⁶⁾	MIN	TYP	MAX	UNIT
PS1	PCB routing plus plane layers		10		
PS2	Signal routing layers		6		
PS3	Full VSS reference layers under DDR routing region ⁽¹⁾	1			
PS4	Full VDDSD_DDR power reference layers under the DDR routing region ⁽¹⁾	1			
PS5	Number of reference plane cuts allowed within DDR routing region ⁽²⁾			0	
PS6	Number of layers between DDR routing layer and reference plane ⁽³⁾			0	
PS7	PCB routing feature size (ie. Copper traces, pads, conductive planes, etc)		4		Mils
PS8	PCB trace width, w		4		Mils
PS9	Point-to-Point, single-ended impedance		40	51	Ω
PS10	Point-to-Point, differential impedance		80	90	Ω
PS11	T-Branch, single-ended impedance ⁽⁵⁾		35/70		Ω
PS12	T-branch, differential impedance ⁽⁵⁾		70/140		Ω
PS13	Impedance control ⁽⁴⁾	Z-10%	Z	Z+10%	Ω

- (1) Ground reference layers are preferred over power reference layers. Return signal vias need to be near layer transitions.
- (2) No traces should cross reference plane cuts within the DDR routing region. High-speed signal traces crossing reference plane cuts create large return current paths, which can lead to excessive crosstalk and EMI radiation. Beware of reference plane voids caused by via antipads, as these also cause discontinuities in the return current path.
- (3) Reference planes are to be directly adjacent to the signal layer, to minimize the size of the return current loop.
- (4) Z is the nominal singled-ended or differential impedance selected for the PCB specified by PS9-PS12.
- (5) Balanced T traces (also referred to as T-branch traces) are split traces from source to multiple end points. The target impedance of the split trace should be 2 times the non-branched impedance. See routing topologies. The maximum trace impedance is typically limited by the minimum achievable trace width. Reduction of the non-branched impedance may be necessary to maintain the 2:1 trace impedance ratio.
- (6) These specifications are to be used as a starting point for designs. It is recommended each design be extracted and simulated to ensure all requirements are met.

1.5 Bypass Capacitors

It is recommended to follow TI's EVM for the decoupling capacitor count and selection. The sections that follow provide some general guidance for decoupling, but recommendation is to follow the EVM. This is because the DDR has been tested/validated on the EVM. Power decoupling also include PCB design elements, so decoupling changes may be required to account for PCB differences.

1.5.1 Bulk Bypass Capacitors

Bulk bypass capacitors are required for moderate speed bypassing of the DDR SDRAMs and other circuitry.

[Table 1-2](#) contains the minimum numbers and capacitance required for the bulk bypass capacitors. [Table 1-2](#) only covers the bypass needs of the processor DDR PHY. Additional bulk bypass capacitance may be needed for other circuitry. For any additional decoupling requirements for the SDRAM devices, see the manufacturer's data sheet.

Table 1-2. Bulk Bypass Capacitors

Number	Parameter	MIN	MAX	UNIT
1	VDDS_DDR bulk bypass capacitor count ⁽¹⁾	1 ⁽²⁾		Devices
2	VDDS_DDR bulk bypass total capacitance	10 ⁽²⁾		μF
3	VDDS_DDR_BIAS bypass capacitor count ⁽¹⁾	1 ⁽²⁾		Devices
4	VDDS_DDR_BIAS bulk bypass total capacitance	10 ⁽²⁾		μF

- (1) These devices should be placed near the devices they are bypassing, but preference should be given to the placement of the high-speed (HS) bypass capacitors and DDR signal routing.
- (2) The capacitor recommendations in this guide reflect only the needs of this processor. For determining the appropriate decoupling capacitor arrangement for the memory device itself, see the memory vendor's guidelines.

1.5.2 High-Speed Bypass Capacitors

High-speed (HS) bypass capacitors are critical for proper DDR interface operation. It is particularly important to minimize the parasitic series inductance of the HS bypass capacitors to VDDS_DDR and the associated ground connections. [Table 1-3](#) contains the specification for the HS bypass capacitors and for the power connections on the PCB. Generally speaking, TI recommends:

- Fitting as many HS bypass capacitors as possible.
- Minimizing the distance from the bypass capacitor to the pins and balls being bypassed.
- Using the smallest physical sized ceramic capacitors possible with the highest capacitance readily available.
- Connecting the bypass capacitor pads to their vias using the widest traces possible and using the largest via hole size possible.
- Minimizing via sharing. Note the limits on via sharing shown in [Table 1-3](#).
- Using three-terminal capacitors in place of two-terminal capacitors. Three-terminal capacitors have lower package inductance and can provide overall lower loop inductance for power delivery. One three-terminal capacitor can often replace multiple two-terminal capacitors.

For any additional SDRAM requirements, see the manufacturer's data sheet.

Table 1-3. High-Speed Bypass Capacitors

Number	Parameter	MIN	TYP	MAX	UNIT
1	HS bypass capacitor package size ⁽¹⁾		0201	0402	Mils
2	Distance, HS bypass capacitor to processor being bypassed ^{(2) (3) (4)}			400	Mils
3	Processor HS bypass capacitor count per VDDSD_DDR rail	12			Devices
4	Processor HS bypass capacitor total capacitance per VDDSD_DDR rail	3.7			μF
5	Number of connection vias for each device power/ground ball	1			Vias
6	Trace length from processor power/ground ball to connection via ⁽²⁾		35	70	Mils
7	Distance, HS bypass capacitor to DDR device being bypassed ⁽⁵⁾			150	Mils
18	DDR device HS bypass capacitor count ⁽⁶⁾	12			Devices
19	DDR device HS bypass capacitor total capacitance ⁽⁶⁾	0.85			μF
10	Number of connection vias for each DDR device power/ground ball	1			Vias
11	Trace length from DDR device power/ground ball to connection via ^{(2) (8)}		35	60	Mils
12	Number of connection vias for each HS capacitor ^{(7) (8)}	2			Vias
13	Trace length from bypass capacitor to connection via ^{(2) (8)}		35	100	Mils

(1) LxW, 10-mil units, that is, a 0402 is a 40 x 20-mil surface-mount capacitor.

(2) Closer/shorter is preferable.

(3) Measured from the nearest processor power or ground ball to the center of the capacitor package.

(4) Five of these capacitors should be located underneath the processor, among the cluster of VDDSD_DDR balls. Two of these capacitors should be located underneath the processor, among the cluster of VDDSD_DDR_BIAS balls.

(5) Measured from the DDR device power or ground ball to the center of the capacitor package. Refer to the guidance from the SDRAM manufacturer.

(6) Per DDR device. For more information, see the guidance from the SDRAM manufacturer.

(7) An additional HS bypass capacitor can share the connection vias only if it is mounted on the opposite side of the board. No sharing of vias is permitted on the same side of the board.

(8) An HS bypass capacitor may share a via with a DDR device mounted on the same side of the PCB. A wide trace should be used for the connection, and the length from the capacitor pad to the DDR device pad should be less than 150 mils.

1.6 Velocity Compensation

For a PCB design, portions of the DDR signal traces are microstrip (BGA break-out segments), but majority of the trace segments are stripline (internal layers). Even though there is a wide variation in the proportion of track length routed as microstrip or stripline, the length/delay matching process should include a mechanism for compensating for the velocity delta between these two types of PCB interconnects. A compensation factor of 1.1 has been specified for this purpose by JEDEC. All microstrip segment lengths are to be divided by 1.1 before summation into the length matching equation. The resulting compensated length is termed the 'stripline equivalent length'. While some amount of residual velocity mismatch skew remains in the design, the process is a substantial improvement over simple length matching.

2 LPDDR4 Board Design and Layout Guidance

2.1 LPDDR4 Introduction

LPDDR4 is an SDRAM device specification governed by the JEDEC standard JESD209-4, Low Power Double Data Rate 4 (LPDDR4). This standard strives to reduce power and improve signal integrity by implementing a lower voltage I/O power rail, employing ODT on the Command/Address bus, and reducing the overall width of the Command/Address bus, among other features. Unlike other DDR types, LPDDR4 has been organized into 2 × 16-bit channels. ECC is supported inline, thus a dedicated SDRAM for ECC is not required.

LPDDR4X is a variant of LPDDR4, with the difference of additional power savings by reducing the I/O voltage from 1.1 V to 0.6 V. It is possible LPDDR4 and LPDDR4X may be available in different packages and/or densities (outside scope of this document). LPDDR4X is currently not supported. It is possible support will be added at a future date, once additional validation efforts and data are collected.

The maximum supported rows for LPDDR4 devices is 17. The JEDEC standard was ratified in 2020, and increased the max number of rows from 17 to 18. As a result, certain high density parts that use byte-mode die and require 18 row bits are not supported.

ECC is supported on the LPDDR4 interface. Unlike traditional ECC interfaces which require dedicated memory pins and devices, ECC is supported inline. The ECC system impact is in interface bandwidth and overall memory density, as ECC data is stored alongside non-ECC data.

For increased memory bandwidths, some device may support multiple LPDDR interfaces. For these devices, the LPDDR interfaces (DDRSS0, DDRSS1, DDRSS2, and so forth) should always be used in incrementing order. For example, if a single LPDDR component is used it should be connected to DDR0_* interface. If two LPDDR components are used they should be connected to DDR0_* and DDR1_* interfaces. Three interfaces should use DDR0_*, DDR1_*, DDR2_*.

The following sections detail the routing specification and layout guidelines for an LPDDR4 interface.

2.2 LPDDR4 Device Implementations Supported

There are different LPDDR4 SDRAM combinations supported by the DDR subsystem. [Table 2-1](#) lists the supported device combinations.

Table 2-1. Supported LPDDR4 SDRAM Combinations

LPDDR4 SDRAM Count	Channels	Die	Ranks	LPDDR4 Channel Width	DDRSS Data Width
1 (2)	2	1	1	16 bits	32 bits
1 (1) (4)	1	1	1	16 bits	16 bits
1 (2)	2	2	1	16 bits	32 bits
1 (3)	2	4	2	16 bits	32 bits

- (1) 16 bit DDRSS data width implementations are non-standard, but supported on select devices. See device-specific data manual if supported for a particular processor.
- (2) See 32-Bit, Single Rank LPDDR4 Implementation.
- (3) See 32-Bit, Dual-Rank LPDDR4 Implementation.
- (4) See 16-Bit, Single Rank LPDDR4 Implementation.

2.3 LPDDR4 Interface Schematics

The LPDDR4 interface schematics vary, depending upon the number of ranks implemented. General connectivity is straightforward and consistent between the implementations. [Figure 2-1](#) illustrates a 32-bit, single-rank LPDDR4 implementation. If dual rank is required, the additional chip select is included. [Figure 2-2](#) illustrates a 32-bit, dual-rank LPDDR4 implementation. On select devices, 16-bit single-rank LPDDR4 implementation are support, see [Figure 2-3](#).

Note

Though LPDDR4 SDRAMs pin out two separate channels, independent channel use is not supported by this processor.

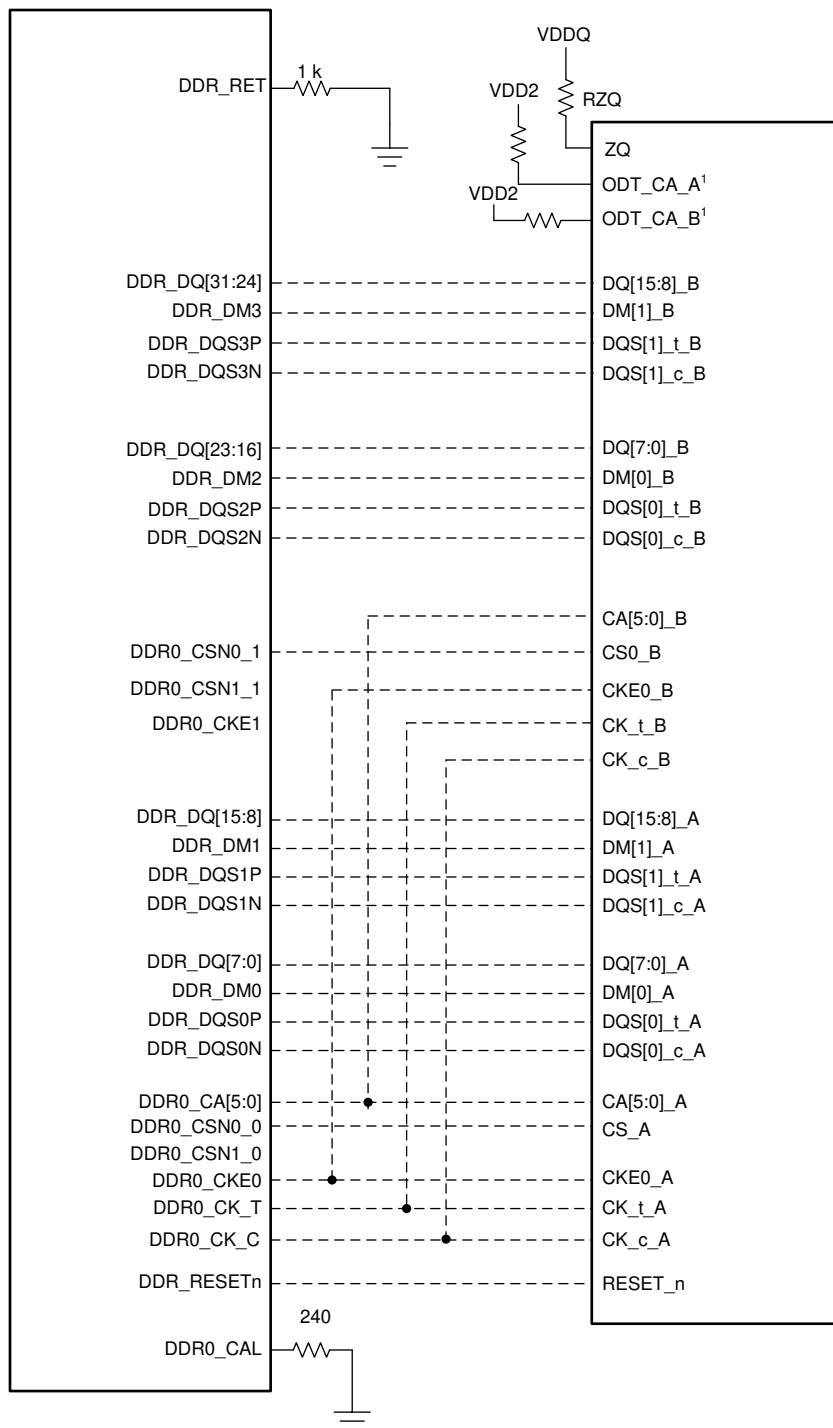


Figure 2-1. 32-Bit, Single-Rank LPDDR4 Implementation

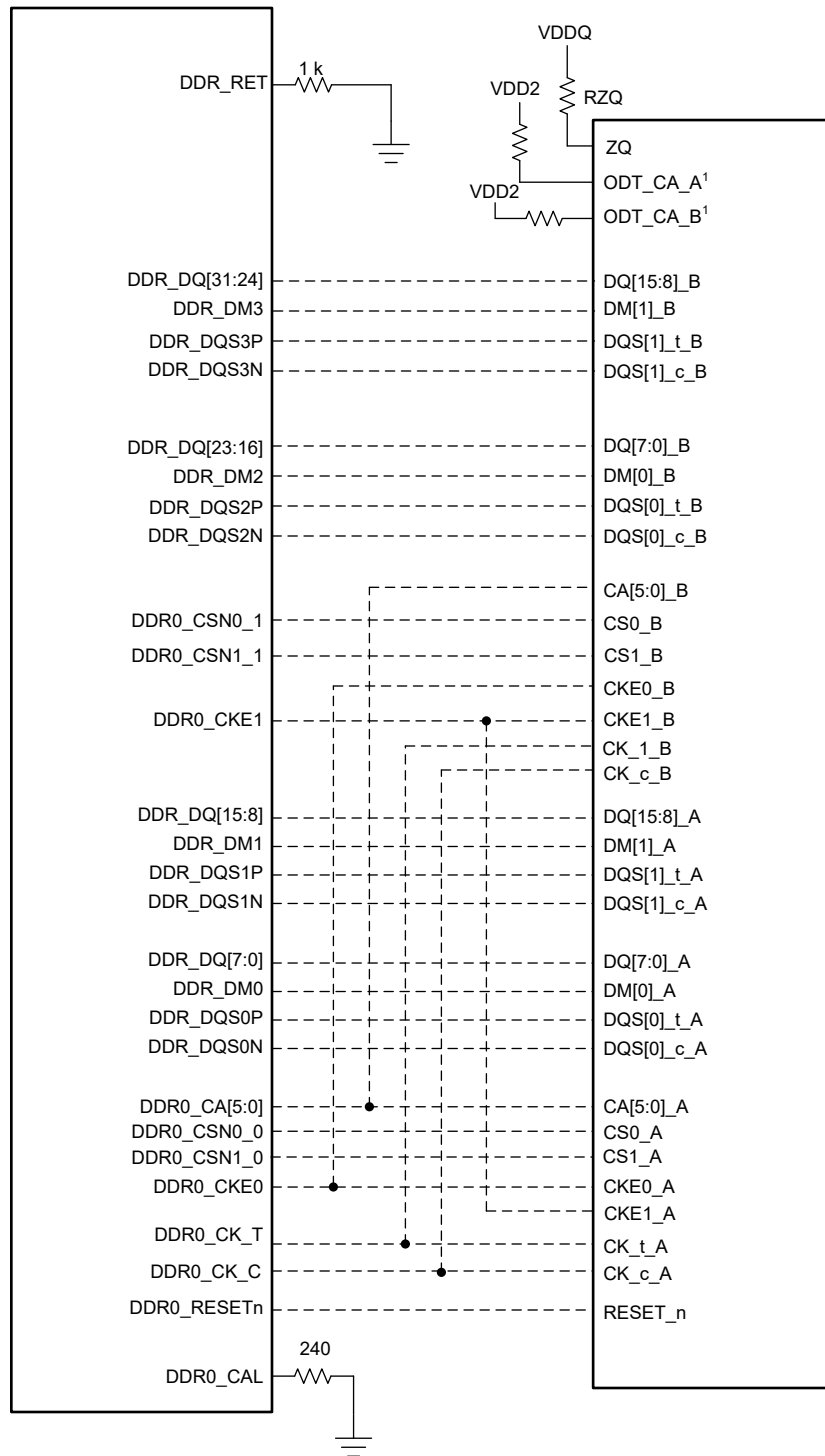


Figure 2-2. 32-Bit, Dual Rank LPDDR4 Implementation

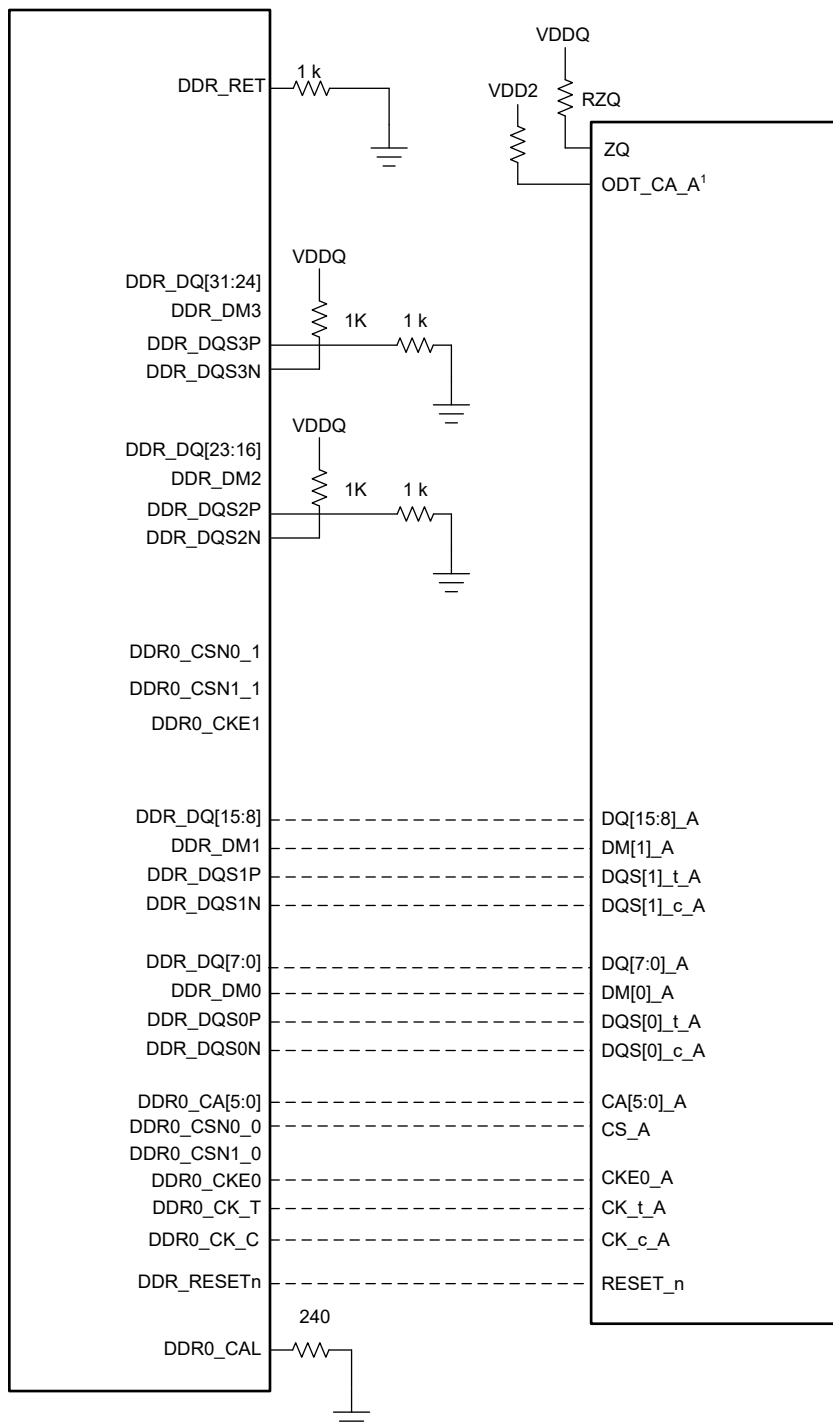


Figure 2-3. 16-Bit, Single Rank LPDDR4 Implementation

2.4 Compatible JEDEC LPDDR4 Devices

Table 2-2 shows the parameters of the JEDEC LPDDR4 devices compatible with this interface.

Table 2-2. Compatible JEDEC LPDDR4 Devices

Number	Parameter	MIN	MAX	UNIT
1	Data Rate ^{(1) (2)}		4266	MT/s
2	Channel Bit Width	x16	x16	Bits
3	Channels	1	2	-
4	Ranks	1	2	-
5	Die	1	4	-
6	Device Count	1	1	-

- (1) For supported data rates, see the device-specific data manual.
(2) SDRAMs in faster speed grades can be used, provided they are properly configured to operate at the supported data rates. Faster speed grade SDRAMs may have faster edge rates, which may affect signal integrity. SDRAMs with faster speed grades must be validated on the target board design.

2.5 Placement

Figure 2-4 shows an example placement for the TDA4VM/DRA829 processor and the LPDDR4 memory device. The recommended spacing parameters for all Jacinto7 devices and the LPDDR4 devices are defined in Table 2-3. The placement does not restrict the side of the PCB on which the devices are mounted. The ultimate purpose of the placement is to limit the maximum trace lengths and allow for proper routing space.

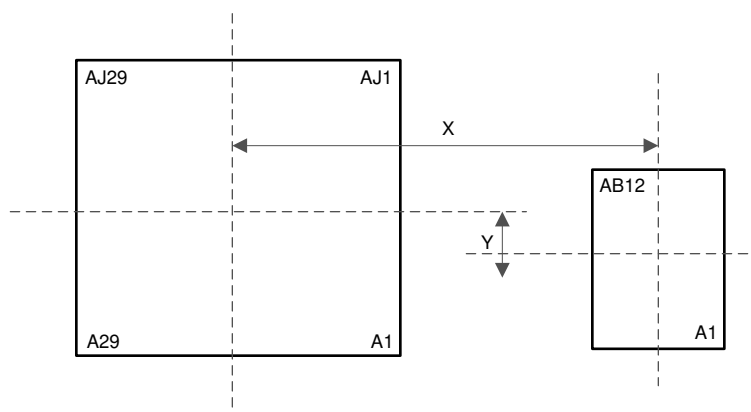


Figure 2-4. TDA4VM/DRA829 LPDDR4 Example Placement (Top View)

Table 2-3. LPDDR4 Placement Recommendations

Number	Parameter	MIN	MAX	UNIT
1	X		1200	Mils
2	Y		250	Mils

2.6 LPDDR4 Keepout Region

The region of the PCB used for LPDDR4 circuitry must be isolated from other signals. The LPDDR4 keepout region is defined for this purpose and example is shown in Figure 2-5. The size of this region varies with the placement and DDR routing. Non-LPDDR4 signals should not be routed on the DDR signal layers within the LPDDR4 keepout region. Non-LPDDR4 signals may be routed in this region only if they are routed on other layers separated from the DDR signal layers by a ground layer. No breaks are allowed in the reference ground layers in this region. In addition, a solid VDDSDDR power plane should exist across the entire keepout region.

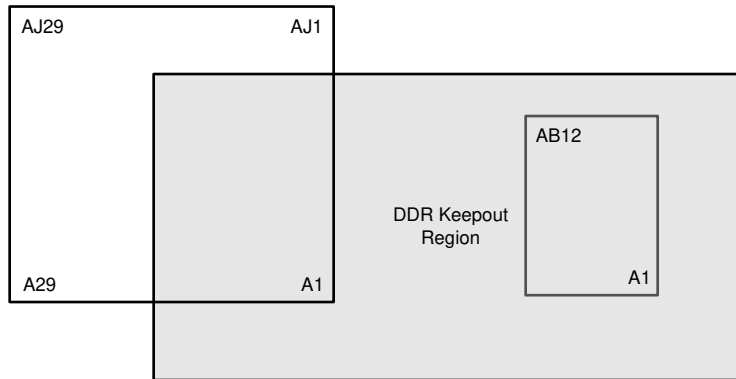


Figure 2-5. Example LPDDR4 Keepout Region for TDA4VM/DRA829 (Top View)

2.7 Net Classes

Routing rules are applied to signals in groups called net classes. Each net class contains signals with the same routing requirements. This simplifies the implementation and compliance of these routes. Table 2-4 lists the clock net classes for the LPDDR4 interface. Table 2-5 lists the signal net classes, and associated clock net classes, for signals in the LPDDR4 interface. These net classes are then linked to the termination and routing rules that follow.

Table 2-4. Clock Net Class Definitions

Clock Net Class	Processor Pin Names
CK	DDR0_CKP / DDR0_CKN
DQS0	DDR0_DQS0P / DDR0_DQS0N
DQS1	DDR0_DQS1P / DDR0_DQS1N
DQS2	DDR0_DQS2P / DDR0_DQS2N
DQS3	DDR0_DQS3P / DDR0_DQS3N

Table 2-5. Signal Net Class Definitions

Signal Net Class	Associated Clock Net Class	Processor Pin Names
CMD_ADDR	CK	DDR0_CA[5:0]
CTRL	CK	DDR0_CS[1:0]_0, DDR0_CS[1:0]_1, DDR0_CKE0, DDR0_CKE1
BYTE0	DQS0	DDR0_DQ[7:0], DDR0_DM0
BYTE1	DQS1	DDR0_DQ[15:8], DDR0_DM1
BYTE2	DQS2	DDR0_DQ[23:16], DDR0_DM2
BYTE3	DQS3	DDR0_DQ[31:24], DDR0_DM3

2.8 LPDDR4 Signal Termination

LPDDR4 memories have software configurable on-die termination for both the data group nets. The DDR subsystem also contains software configurable on-die termination for the address/control group nets. Thus, termination is not required on any DDR signals for an LPDDR4 configuration.

2.9 LPDDR4 VREF Routing

LPDDR4 memories generate their own VREFCA and VREFDQ internally for the address / command bus and data bus, respectively. Similarly, the DDR PHY also provides its own reference voltage for the data group nets during reads. Thus unlike DDR3 and DDR4, VREF does not need to be generated on the board, and there is no required VREF routing for an LPDDR4 configuration.

2.10 LPDDR4 VTT

Unlike DDR3 and DDR4, there is no required termination on the PCB of the address/control bus of an LPDDR4 configuration. All termination is handled internally (on-die). Thus, VTT does not apply for LPDDR4.

2.11 CK, CMD_ADDR, and CTRL Topologies

The CK, CMD_ADDR, and CTRL net classes are routed similarly, and are skew matched from the DDR controller in the processor to the LPDDR4 SDRAM to ensure that the signals are properly sampled at the SDRAM. The CK net class requires more care because it runs at a higher transition rate and are differential. The CK and CMD_ADDR trace routing topology is a balanced 'T' while the CTRL trace routing topology is mix of point-to-point and balanced 'T'.

Figure 2-6 shows the topology of the CK0 net class. Figure 2-7 and Figure 2-8 shows the topologies for the corresponding CMD_ADDR and CTRL net class. Note some of the signals within the group are shared between the memory channels, while other signals are dedicated for each channel. Skew matching requirements for the routing segments are detailed in Table 2-6.

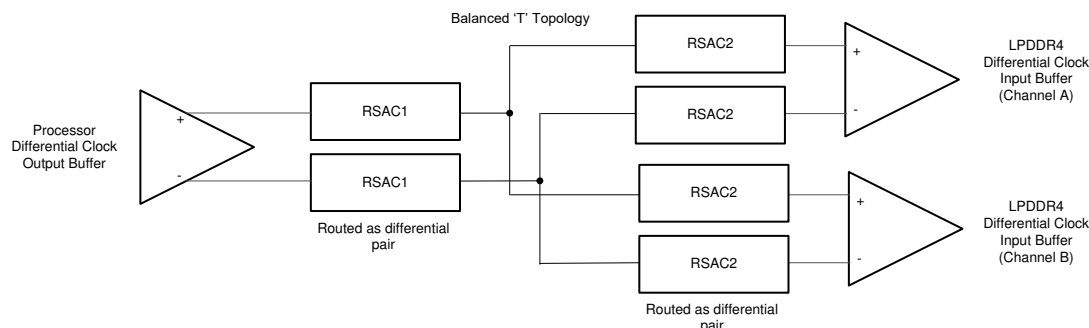


Figure 2-6. LPDDR4 CK Topology

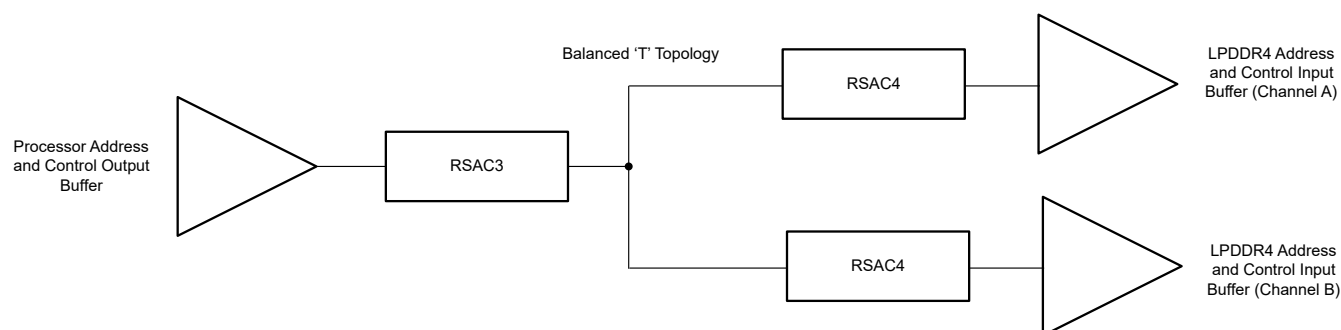


Figure 2-7. LPDDR4 CMD_ADDR, CTRL Topology/Branched Signals

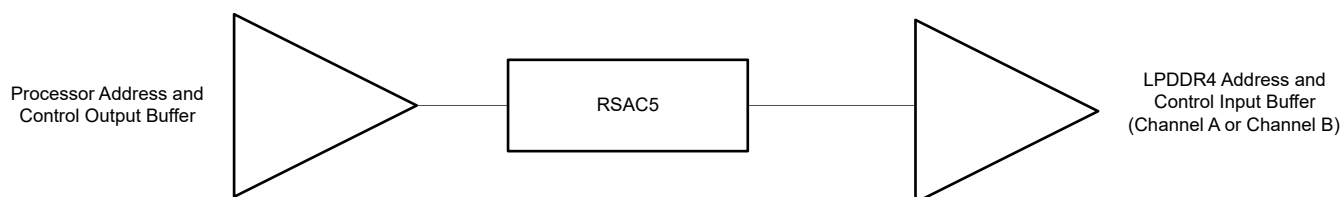


Figure 2-8. LPDDR4 CTRL Topology/Point-to-Point Signals

It is recommended to minimize layer transitions during routing. If a layer transition is necessary, it is preferable to transition to a layer using the same reference plane. If this cannot be accommodated, ensure there are nearby stitching vias to allow the return currents to transition between reference planes. The goal is to minimize the size of the return current path thus minimizing the inductance in this path. Lack of these stitching vias results in impedance discontinuities in the signal path that increase crosstalk and signal distortion.

There are no stubs or termination allowed on the nets of the CK, CMD_ADDR, and CTRL group topologies. All test and probe access points must be in line without any branches or stubs.

2.12 Data Group Topologies

The data line topology is always point-to-point for LPDDR4 implementations, and is broken up into four different byte lanes. It is recommended to minimize layer transitions during routing. If a layer transition is necessary, it is better to transition to a layer using the same reference plane. If this cannot be accommodated, ensure there are nearby ground vias to allow the return currents to transition between reference planes. The goal is to provide a low inductance path for the return current. To optimize the skew matching, TI recommends routing all nets within a single data routing group on one layer where all nets have the exact same number of vias and the same via barrel length.

DQSP and DQSN lines are point-to-point signals routed as a differential pair. [Figure 2-9](#) illustrates the DQSP/N connection topology.

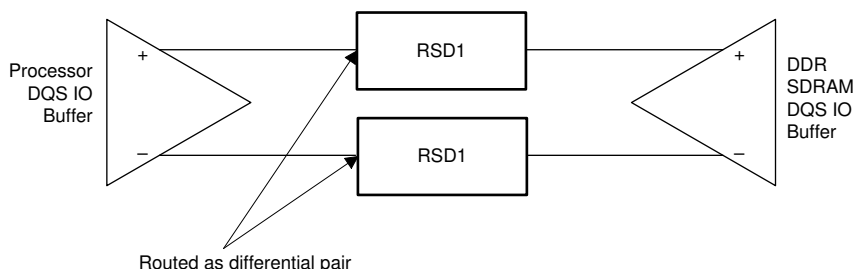


Figure 2-9. LPDDR4 DQS Topology

DQ and DM lines are point-to-point signals routed as single-ended. [Figure 2-10](#) illustrates the DQ and DM connection topology.

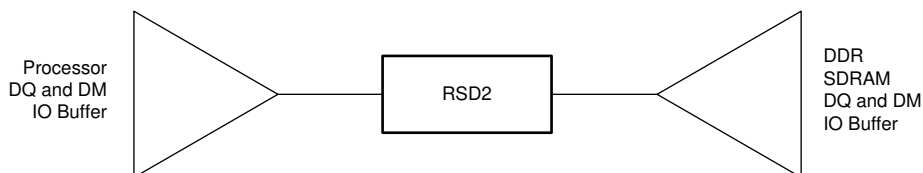


Figure 2-10. LPDDR4 DQ/DM Topology

There are no stubs or termination allowed on the nets of the data group topologies. All test and probe access points must be in line without any branches or stubs.

2.13 CK, CMD_ADDR, and CTRL Routing Specification

Skew within the CK, CMD_ADDR, and CTRL net classes can reduce setup and hold margin for the signals at the SDRAM device. Thus, this skew must be controlled. The routed PCB track has a delay proportional to its length. Thus, the delay skew must be managed through matching the lengths of the routed tracks within a defined group of signals. The only way to practically match skew on a PCB is to lengthen the shorter traces up to the length of the longest net in the net class and its associated clock. Make sure to include the Z-axis delays (vias) during analysis.

Table 2-6 lists the limits for the individual segments that comprise the routing from the processor to the SDRAM. These segment lengths coincide with the CK, CMD_ADDR, and CTRL topology diagrams shown previously in Table 2-6, Figure 2-7, and Figure 2-8. By controlling the routed lengths for the same segments of all signals in a routing group, the signal delay skews are controlled. Most PCB layout tools can be configured to generate reports to assist with this validation. If this cannot be generated automatically, this must be generated and verified manually.

These parameters are recommendations only, intended to get the design close to success prior to simulation. To ensure the PCB design meets all requirements, it is required the design be simulated and those results compared with the simulation results defined in Section 3.

Table 2-6. CK, CMD_ADDR, and CTRL Routing Specifications

Number	Parameter	MIN	TYP	MAX	UNIT
LP4_ACRS1	Propagation delay of net class CK RSAC1 + RSAC2 RSAC1 = CK			300 ⁽¹⁾	ps
LP4_ACRS2	Propagation delay of net class CMD_ADDR and CTRL RSAC3 + RSAC4, RSAC5 RSAC3 = CA RSAC5 = CSN			300 ⁽¹⁾	ps
LP4_ACRS3	Skew within net class CK (CK+ to CK- Skew) (RSAC1 + RSAC2) Skew Kolla upp detta! ⁽⁹⁾			0.75 ⁽²⁾	ps
LP4_ACRS4a	Skew across net class CMD_CTRL RSAC3 + RSAC4 Skew ⁽⁹⁾ CA har mycket bra avstånd!		10	30	ps
LP4_ACRS4b	Skew across net class CTRL RSAC3 + RSAC4, RSAC5 Skew ⁽⁹⁾ CSN har mycket bra avstånd!		10	30	ps
LP4_ACRS5	Skew between each T-branch signal pair RSAC2 or RSAC4 Skew ⁽⁷⁾ Använder inte T-branch!		0.1		ps
LP4_ACRS6	Skew across CMD_ADDR, and CTRL and associated CK clock net class RSAC1+ RSAC2, RSAC3 + RSAC4, RSAC5 ⁽⁹⁾ Jag måste öka CA och CK!!!			60	ps
LP4_ACRS7	Vias per trace Använder inga vior			4	vias
LP4_ACRS8	Via Stub Length ⁽⁸⁾ Använder stub length 1.55mm som max		20		Mils
LP4_ACRS9	Via count difference			0 ⁽³⁾	vias
LP4_ACRS10	Center-to-center CK to other LPDDR4 trace spacing Stor avstånd!	4w ⁽⁴⁾			
LP4_ACRS11	Center-to-center CMD_ADDR, CTRL to other LPDDR4 trace spacing	3w ⁽⁴⁾			Stort avstånd!
LP4_ACRS12	Center-to-center CMD_ADDR, CTRL to self or other CMD_ADDR, CTRL trace spacing	3w ⁽⁴⁾			Stort avstånd!
LP4_ACRS13	CK center-to-center spacing ^{(5), (6)}				
LP4_ACRS14	CK spacing to other non-DDR net	4w ⁽⁴⁾			

- (1) Max value is based upon conservative signal integrity approach. This value could be extended only if detailed signal integrity analysis of rise time and fall time confirms desired operation.
- (2) To be verified by design/simulation, confirming JEDEC defined Vix CK_ratio (25%) are satisfied with good eye margins.
- (3) Via count difference may increase by 1 only if accurate 3-D modeling of the signal flight times – including accurately modeled signal propagation through vias – has been applied to ensure all segment skew maximums are not exceeded.
- (4) Center-to-center spacing is allowed to fall to minimum 2w for up to 500 mils of routed length (only near endpoints). Spacing minimums may be relaxed if simulations accurately capture crosstalk between neighboring victim and aggressor traces show good margin. Also consider via spacing. Signals with adjacent vias near SoC should NOT have adjacent vias near DRAM.
- (5) CK spacing set to ensure proper differential impedance.
- (6) The user must control the impedance so that inadvertent impedance mismatches are not created. Generally speaking, center-to center spacing should be either 2w or slightly larger than 2w to achieve a differential impedance equal to twice the single-ended impedance, Zo, on that layer.

- (7) Skew control on branch trace segments (Balance T) is intended to optimize signal integrity (waveform reflections). It is not required nor recommended to match skew across all branch trace segments, just for each branch of a specific signal.
- (8) Via stub length control (micro-via or backdrill) may be required if operating LPDDR4 above 3200 Mbps (depending on simulation).
- (9) Recommend routing net classes CK, CMD_ADDR, and CTRL on same signal layer for better skew control.

2.14 Data Group Routing Specification

Skew within the Byte signal net class directly reduces the setup and hold margin for the DQ and DM nets. As described with the ADDR_CTRL signal net class and associated CK clock net class, this skew must be controlled. The data byte skew must be managed through controlling the lengths of the routed tracks within a defined group of signals. The only way to practically match skews on a PCB is to lengthen the shorter traces up to the length of the longest net in the net class and its associated clock. Make sure to include the Z-axis delays (vias) during analysis.

Note

It is not required nor recommended to match the lengths across all byte lanes. Length matching is only required within each byte.

Table 2-7 contains the routing specifications for the Byte0, Byte1, Byte2, and Byte3 routing groups. Each signal net class and its associated clock net class is routed and matched independently. These parameters are recommendations only, intended to get the design close to success prior to simulation. To ensure the PCB design meets all requirements, it is required the design be simulated and those results compared with the simulation results defined in Section 3.

Table 2-7. Data Group Routing Specifications

Number	Parameter	MIN	TYP	MAX	UNIT
LP4_DRS1	Propagation delay of net class DQSx RSD1 ⁽⁴⁾ RSD1 = DQS			300 ⁽²⁾	ps
LP4_DRS2	Propagation delay of net class BYTEx RSD2 RSD2 = DQ			300 ⁽²⁾	ps
LP4_DRS3	Propagation delay of each DQS pair must be less than propagation delay of CK pair. RSD1 < (RSAC1 + RSAC2) ⁽⁴⁾ DQS är inte kortare än CK!	0			ps
LP4_DRS4	Skew within net class DQSx RSD1 Skew (DQS+ to DQS-) ^{(4) (5)}			0.75	ps
LP4_DRS5	Skew across net class DQSx and BYTEx RSD1 to RSD2 Skew ^{(1), (4), (5)} Inga problem här!		5	25	ps
LP4_DRS6	Propagation delay of each DQS pair must be less than propagation delay DQ/DM. RSD1 < RSD2 ⁽⁴⁾ Kolla upp detta!	0		150 ps	ps
LP4_DRS7	Vias Per Trace			2 ⁽²⁾	vias
LP4_DRS8	Via Stub Length ⁽³⁾		40		Mils
LP4_DRS9	Via Count Difference			0 ⁽⁶⁾	vias
LP4_DRS10	RSD1 center-to-center spacing (between clock net class) ⁽⁷⁾	4w			
LP4_DRS11	RSD1 center-to-center spacing (within clock net class) ^{(8), (9)}				
LP4_DRS12	RSD2 center-to-center spacing (between signal net class) ⁽⁷⁾	4w			
LP4_DRS13	RSD2 center-to-center spacing (within signal net class) ⁽⁷⁾	3w			

- (1) Skew matching is only done within a byte. Skew matching across bytes is neither required nor recommended.
- (2) Max value is based upon conservative signal integrity approach. This value could be extended only if detailed signal integrity analysis of rise time and fall time confirms desired operation.
- (3) Via stub length control (micro-via or backdrill) may be required if operating LPDDR4 above 3200 Mbps (depending on simulation).
- (4) Consider the delays from SoC die pad to the DRAM pin (ie. Delays include SoC package plus PCB). Consider only one leg of any T-branch trace segments.
- (5) To be verified by design/simulation, confirming JEDEC defined Vix_DQS_ratio (20%) are satisfied with good eye margins.
- (6) Via count difference may increase by 1 only if accurate 3-D modeling of the signal flight times – including accurately modeled signal propagation through vias – has been applied to ensure DQn skew and DQSn to DQn skew maximums are not exceeded.
- (7) Center-to-center spacing is allowed to fall to minimum 2w for up to 500 mils of routed length (only near endpoints). Spacing minimums may be relaxed if simulations accurately capture crosstalk between neighboring victim and aggressor traces show good margin. Also consider via spacing. Signals with adjacent vias near SoC should NOT have adjacent vias near DRAM.
- (8) DQS pair spacing is set to ensure proper differential impedance.
- (9) The Simulation Results must control the impedance so that inadvertent impedance mismatches are not created. Generally speaking, center-to-center spacing should be either 2w or slightly larger than 2w to achieve a differential impedance equal to twice the single-ended impedance, Zo, on that layer.

2.15 Channel, Byte, and Bit Swapping

All signals, including data and address/control, must be routed 1 to 1 from the DDR controller to the LPDDR4 memory. Byte swapping across channels or within a channel is not allowed. Similarly, data bit swapping across byte lanes or within a byte is also not allowed. In addition, byte lanes 0 and 1 of the DDR controller must be routed to channel A of the LPDDR4 memory, and byte lanes 2 and 3 of the DDR controller must be routed to channel B of the LPDDR4 memory.

3 LPDDR4 Board Design Simulations

This section is intended to provide an overview of the basic system-level board extraction, simulation, and analysis methodologies for high-speed LPDDR4 interfaces. This is an essential step to ensure the PCB design meets all the requirements to operate the targeted speeds.

3.1 Board Model Extraction

The board level extraction guidelines listed below are intended to work in any EDA extraction tool and are not tool-specific. It is important to follow the steps outlined in [Section 3.2](#) through [Section 3.4](#) immediately after completing touchstone model extractions. The design should be checked with these steps prior to running IBIS simulations.

1. For DDR extractions, extract power (VDDS_DDR/VDDQ and VDDS_DDR_BIAS/VDDQX) and signal nets together in a 3D-EM solver. For SerDes extractions power is not required as the simulations are only intended for Signal Integrity
2. Use wide-band models. It is recommended to extract from DC to at least till 6x the Nyquist frequency (for example, for LPDDR4-4266 extract the model at least till 12.8 GHz).
3. Check the board stack-up for accurate layer thickness and material properties.
 - a. It is recommended to use Djordjevic-Sarkar models for the dielectric material definition.
4. Use accurate etch profiles and surface roughness for the signal traces across all layers in the stack-up.
5. If the board layout is cut prior to extraction (to reduce simulation time), please define a cut boundary that is at least 0.25 inch away from the signal and power nets.
6. Check the via padstack definitions.
 - a. Ensure that the non-functional internal layer pads on signal vias are modeled the same way they would be fabricated.
 - b. These non-functional internal layer pads on signal vias are not recommended by TI.
7. Use Spice/S-parameter models (typically available from the vendor) for modeling all passives in the system

3.2 Board-Model Validation

The extracted board models need to be checked for the following properties:

- Passivity: This ensures that the board model is a passive network and does not generate energy
- Causality: This ensures that the board model obeys the causal relationship (output follows input).

These checks can be performed in any standard EDA simulator or extraction engine.

3.3 S-Parameter Inspection

Once the extracted S-parameters have been verified as causal and passive, the S-parameter plots should be inspected. It is recommended to check for the following:

- Insertion Loss: The single-ended insertion loss is recommended to stay within 0 to 10 dB up to 3 times the Nyquist frequency of operation. For example, if the target frequency is 8 Gbps (4GHz Nyquist), the single-ended insertion loss should stay under 10 dB up to 12 GHz.
- Return Loss: The single-ended return loss is recommended to be less than 15 dB up to 3 times the Nyquist frequency.
- Near and Far end crosstalk (FEXT/NEXT): The FEXT and NEXT are recommended to be under 25 dB for frequencies up to 3 times the Nyquist frequency.

The S-Parameter inspection plots are not pass/fail tests, but rather its more of a guide to check if the design has a reasonable chance of performing a the required level.

3.4 Time Domain Reflectometry (TDR) Analysis

As a lot of the design fixes are targeted towards maintaining uniform trace impedance, an important analysis method used in assessing the quality of the design is the Time Domain Reflectometry (TDR) analysis. This plots the impedance of a trace as a function of its length, as shown in [Figure 3-1](#).



Figure 3-1. TDR Plot Example With Impedance Mismatch

As shown in [Figure 3-1](#), the TDR plot highlights impedance discontinuities in the trace from one end to the other. This method depends on a reflected waveform from the far-end of the trace. The delay in the plot corresponding to a particular point in the trace actually corresponds to 2 times the distance of that point from the source, owing to the round trip time. This needs to be factored in for assessing the source of impedance discontinuities.

The TDR plot can be generated by reading in the S-parameter models generated by the extraction tool and assessing them in “Time-Domain” mode. A standard EDA simulator such as HyperLynx can perform this function. It is recommended to optimize the design to within a $\pm 5\%$ deviation from the nominal trace impedance.

The TDR plots are not pass/fail tests, but rather is more of a guide to check if the design has a reasonable chance of performing at the required level.

3.5 Simulation Integrity Analysis

The methodology for validating the DDR interface is outlined in this section. LPDDR4 interfaces, as defined in the JEDEC specification, uses eye masks defined at a target BER (Bit Error Rate) to determine pass or fail for signal integrity. It is essential to perform channel simulations using IBIS models to generate the signal eye diagrams at the targeted BER. Channel simulation techniques are widely used in SerDes interface analysis. These are introduced for memory interfaces starting from DDR4 and LPDDR4.

3.5.1 Simulation Setup

Set up the system-level schematic in the simulator by connecting the SoC IBIS model, SoC package model, board model, DRAM package model (if available), DRAM IBIS model, and power supplies. A typical system-level DDR schematic is shown in [Figure 3-2](#).

Note

Be aware of the DRAM configuration (number of dies in the package, number of ranks, and number of channels) while setting up the system schematic.

Be aware the DRAM configuration may also include On-Die Decoupling Circuit.

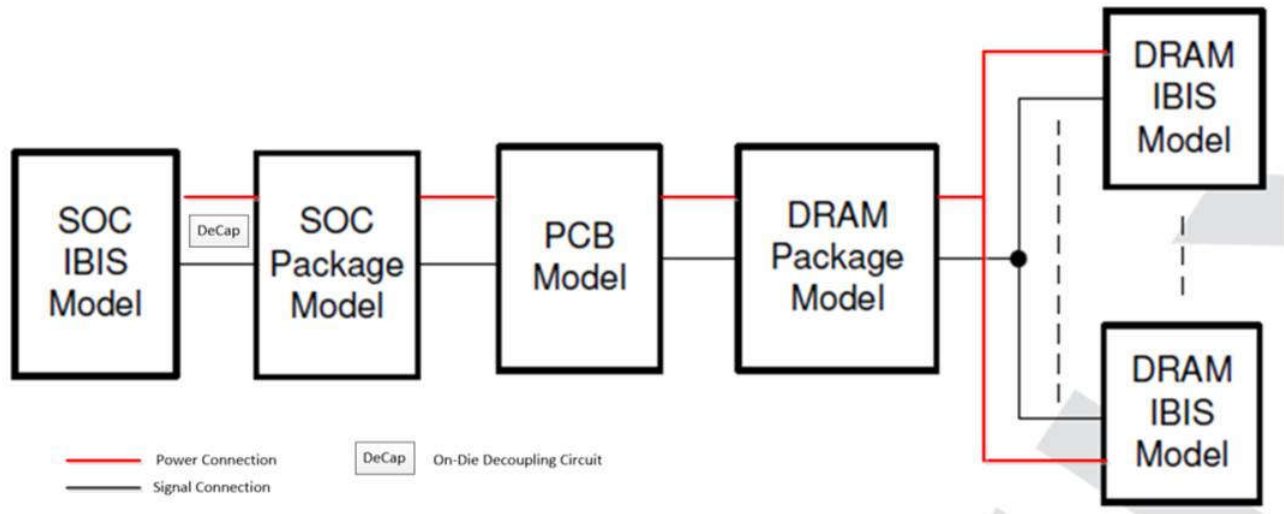


Figure 3-2. Typical System-Level DDR Schematic

- LPDDR4 simulations require power-aware IBIS models for the controller and the memory along with a simulator that supports channel simulations for DDR interfaces.
- SPICE-based transistor-level simulations cannot be used for generating BER signal eyes. Use a simulator that can handle power-aware IBIS simulations and can run channel simulations for the DDR interface.
- IBIS models reduce simulation time with minimal loss in accuracy compared with SPICE-based transistor-level simulations. IBIS models starting from version 5.0 are power-aware models which enables Simultaneous Switching Output (SSO) noise simulations. The TI IBIS model is a power-aware IBIS model.
- Use SPICE models to accurately model the on-die decoupling capacitance on the DDR supply net for both – controller and DRAM. This ensures accurate power noise and Power Supply Induced Jitter (PSIJ) estimation in DDR simulations. The on-die decoupling capacitance information for the DRAM can be obtained from the DRAM vendor.
- Use SPICE or S-parameter files to model the DRAM package. This can be requested from the DRAM vendor. EBD models are not recommended.

- J721E model for the on-die decoupling capacitance on the DDR supply net:

```
*****
* On-die Decoupling circuit for J7ES (DIE_VDDS_DDR to VSS)
*****
* Notes:
* Includes on-die decoupling for all DDR signals
*
* This subcircuit should be added across the J7ES IBIS model
* DIE_VDDS_DDR and VSS pins
*
*****
* x_decouple DIE_VDDS_DDR vss_die J7ES_ondie_decoupling_alldq
*****
.SUBCKT J7ES_ondie_decoupling_alldq DIE_VDDS_DDR vss_die
Cvddq_c DIE_VDDS_DDR DIE_VDDS_DDR_c 2105.86e-12
Rvddq_c vss_die DIE_VDDS_DDR_c 43e-3
.ENDS
```

3.5.2 Simulation Parameters

It is important to configure the simulation to exercise the system to real, but worst case parameters.

- Use the worst-case bit pattern to excite the system. The simulator should be able to generate the worst-case bit pattern based on channel characterization.
- Select the controller and DRAM models (sets the drive strength, ODT, VOH levels, and so forth) from the IBIS files which work best for the system.
 - Selection of DRAM model setting is typically an iterative process.
 - Every system is unique and the optimal settings for these parameters can vary from system to system.

Table 3-1. Example Data Write ODI/ODT Optimization

Pkg Byte	Board	ODI Ω	ODT Ω	Total EW Margin (ps)	Total EH Margin (mV) _B
B3	J7 370HR 10L Ref B3, No BD	40	40	50.28	15.66
B3	J7 370HR 10L Ref B3, No BD	40	48	27.62	11.76
B3	J7 370HR 10L Ref B3, No BD	40	40	33.52	2.92
B3	J7 370HR 10L Ref B3, No BD	48	48	1.54	0.86

- Data bus and address bus ODT and drive strength values can be set independently. As an example, the J721E EVM used 40- Ω ODT for data read/writes and 80- Ω for CA bus. Drive strength of 40-ohms for data read/write and CA.
 - Data READ Controller model - lpddr4_odt_40, lpddr4_odt_40_diff
 - Data WRITE Controller model - lpddr4_ocrd_40p_40n, lpddr4_ocrd_40p_40n_diff
 - CA/CLOCK Controller model - lpddr4_ocrd_40p_40n, lpddr4_ocrd_40p_40n_diff
- Set up the channel simulation parameters. These typically consists of the data rate, ignore time/bits, minimum number of bits, bit sampling rate, BER floor, number of bits for display, types of BER eyes (voltage and/or timing), and target BER.
 - To determine the minimum number of bits one can run a series of channel simulations with different number of bits. The BER signal eye (and margins) tend to converge after a certain minimum number of bits. This should help determining the minimum number of bits to be used for the system.
 - Run channel simulations to generate the eye diagrams at LBER of -16.
- Run channel simulations with non-ideal power settings at different PVT corners. It is recommended to run the simulations at least at the SSHT and FFLT corners.

3.5.3 Simulation Targets

Once the simulation successfully completes, generate the DDR analysis reports from the simulation tool. There are several different parameters to be verified, detailed in this section. Each parameter is pass/fail, meaning each must meet the specified target to ensure the design has sufficient margin to operate at the target data rates.

Use the appropriate JEDEC Vref parameters (Vref_min, Vref_max, Vref_step, and Vref_set_tol) and mask parameters (share, height, width).

3.5.3.1 Waveform Quality

The minimum ring-back margins at low and high levels are required to be verified, as well as a review the peak-peak power noise. Transient analysis should be performed using IBIS models to generate the power noise. Below are examples of measurements from the waveforms.

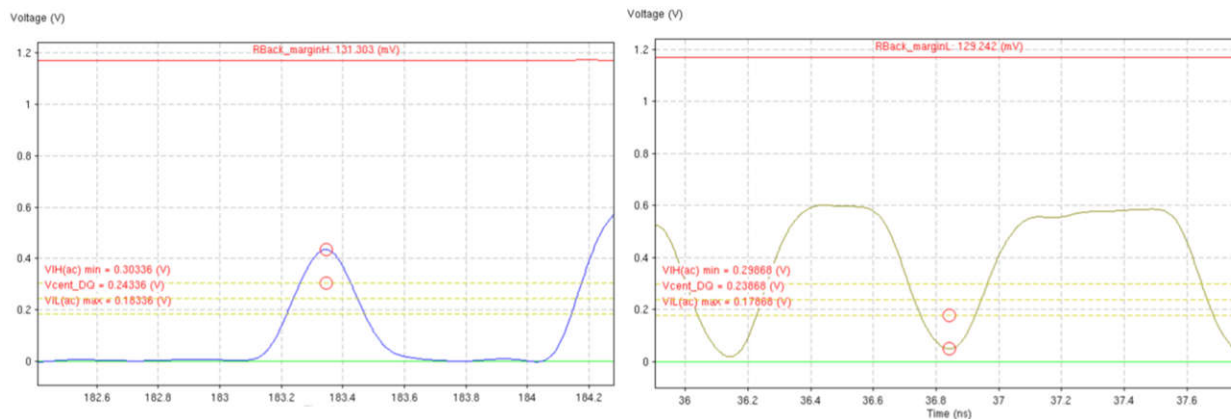


Figure 3-3. Example of Ring-back Margin Waveforms (High/Low)

3.5.3.2 Eye Quality

The Vix_DQS ratio and Vix_CK ratio for data write and CA bus simulations are to be verified, at the DRAM pin/BGA. Figure 3-4 from the JEDEC specification explains how to measure the Vix ratio, as well as define the ration requirement(s).

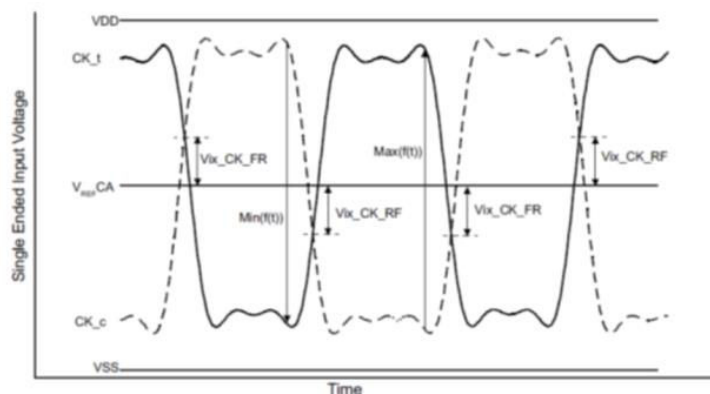


Figure 3-4. Vix_CK and Vix_DQS Ratio for Eye Quality

Table 3-2. Vix_CK and Vix_DQS Ratio for Eye Quality

Symbol	Data Rate						Unit	Note
	1600/1867		2133/2400/3200		3733/4266			
Vix_CK_ratio	-	25	-	25	-	25	%	(1), (2)
Vix_DQS	-	20	-	20	-	20	%	(1), (2)

(1) Vix_CK_Ratio is defined by the equation: $Vix_CK_Ratio = Vix_CK_FR / \text{Min}(f(t))$

(2) Vix_CK_Ratio is defined by the equation: $Vix_CK_Ratio = Vix_CK_RF / \text{Min}(f(t))$

3.5.3.3 Delay Report

The required interconnect delays for DQ, DQS, CA, and CLK are listed in [Table 2-6](#) and [Table 2-7](#). The values listed as 'Typical' are only recommendations. Any minimum/maximum value is a requirement. One key requirement is to ensure the CK delay is greater than any DQS delay.

3.5.3.4 Mask Report

The minimum jitter and noise margins are to be captured with respect to the eye mask(s). This masks are data rate dependent, and includes:

- Data read eye mask at the SOC die pad for functionality testing
- Data write eye mask (JEDEC spec) at the DRAM pin/BGA for compliance testing
- CA bus eye mask (JEDEC spec) at the DRAM pin/BGA for compliance testing

There should be at least 2 sets of eye diagrams generated by the simulator:

- Vref set to the optimal Vref of the byte offset by the Vref_set_tol in the positive direction (Vref_set_tol is defined in JEDEC spec)
- Vref set to the optimal Vref of the byte offset by the Vref_set_tol in the negative direction

The system-level margins are the worst case noise and jitter margins from all eye diagram measurements listed above (across SSHT and FFLT corners). For all waveforms captured at the DRAM device, margins should be calculated at both the BGA pin and the DRAM pad.

Table 3-3. LPDDR4/4x Eye Mask Definitions/Requirements

Parameter	Mask Shape	LPDDR4-3200	LPDDR4-3733	LPDDR4-4266
CA eye mask TcIVW	Rectangular ⁽¹⁾	0.3 UI ⁽¹⁾	⁽²⁾	0.3 UI ⁽¹⁾
CA eye mask VcIVW	Rectangular ⁽¹⁾	155 mV ⁽¹⁾	⁽²⁾	145 mV ⁽¹⁾
Write eye mask TdIVW	Rectangular ⁽¹⁾	0.25 UI ⁽¹⁾	⁽²⁾	0.25 UI ⁽¹⁾
Write eye mask VdIVW	Rectangular ⁽¹⁾	140 mV ⁽¹⁾	⁽²⁾	120 mV ⁽¹⁾
Read eye mask TdIVW	Diamond	0.61 UI	0.66 UI	0.7 UI
Read eye mask VdIVW	Diamond	140 mV	120 mV	120 mV

(1) Copied from JEDEC specification: Low Power Double Data Rate 4 (LPDDR4).

(2) For details, contact the DRAM vendor.

Figure 3-5 through Figure 3-7 show the eye mask definitions translated to eye diagrams within captured waveforms.

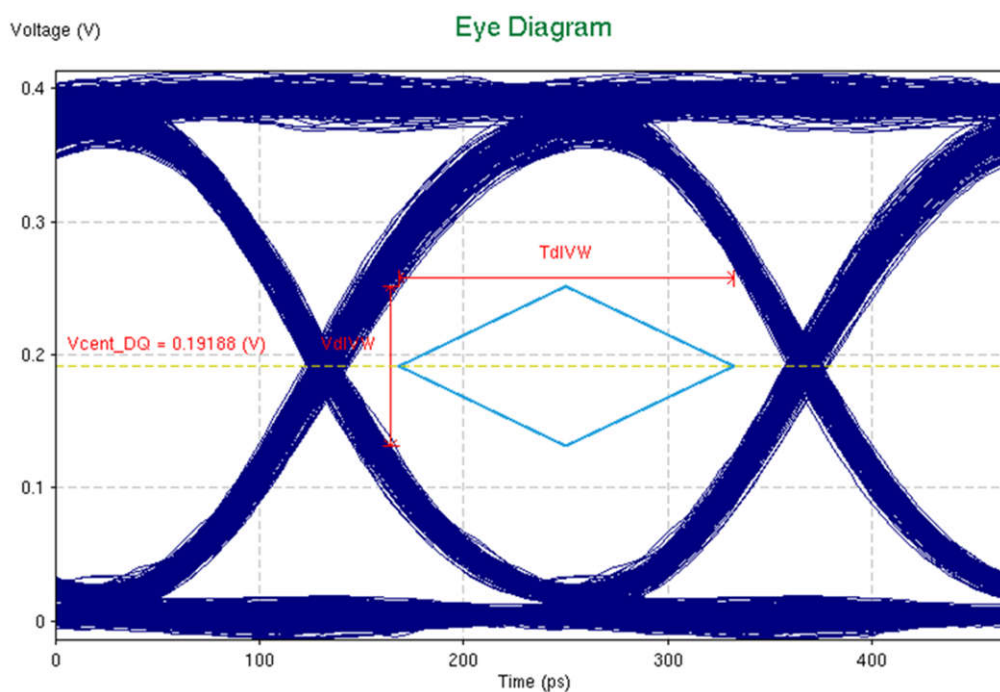


Figure 3-5. Sample Simulated LPDDR4-4266 Read Eye With Diamond-Shaped Eye Mask

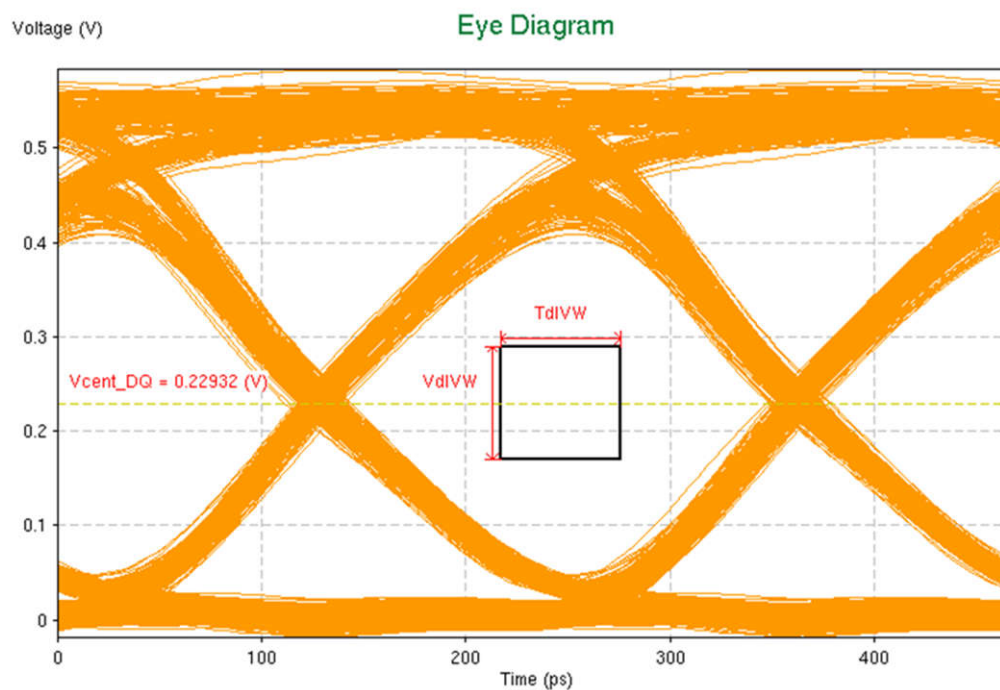


Figure 3-6. Sample Simulated LPDDR4-4266 Write Eye With Rectangular JEDEC Eye Mask

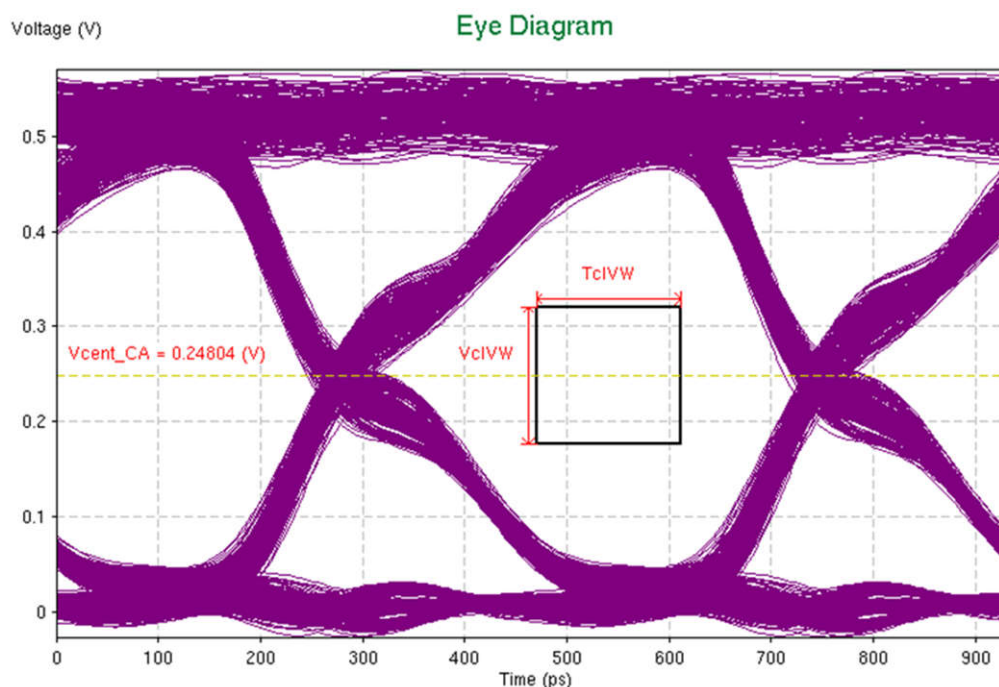


Figure 3-7. Sample Simulated LPDDR4-4266 CA Eye With Rectangular JEDEC Eye Mask

3.6 Design Example

3.6.1 Stack-Up

These guidelines recommend a 10 layer PCB stack-up for full device entitlement. Below is an example stack-up from a 10Lyr reference design.

- Designs using FR4 products like 370HR are supported, but also recommend higher speed materials like ISOLA I-Speed (or equivalent) for increased margin.
- This example routes data groups on layers 2 and 4. While this minimizes the via travel and therefore reduces via-to-via coupling, but it leaves a longer via stub, which might require back-drill.
- Dynamic CA signals are routed on layer 7, and more static control signals routed on layer 9.

Table 3-4. Example PCB Stackup for LPDDR4

Layer No	Stackup	Routing Plan Highest Priorities and Layer
	Solder mask	
1	TOP - PWR/SIG	BGA breakouts/VDD_CPU, VDD_CORE and VDD_DDR_1V1
2	PWR/SIG	VDD_CPU and CORE/LPDDR (DBG #3/#1, CAT-Branched)
3	GND	REF
4	PWR/SIG	VDDA_PHYCORE_0V8, VDD_0V8, 0V85/LPDDR (DBG #2/#0)
5	PWR/GND	VDDA_0V8_0V8 and GND flood for LPDDR4
6	PWR/GND	VDD_0V8, VDDA_0V8 supplies and GND flooded for LPDDR4
7	SIG/PWR	VDD_0V8, VDDA_0V8/LPDDR (Dynamic CA, Trunks)/SERDES
8	GND	REF
9	SIG/PWR	VDD_0V8, VDDA_0V8/LPDDR (static CA)
10	BOTTOM - SIG/PWR	BGA breakouts/Pwr and GND plan segments
	Solder mask	

Table 3-5 provides simulation results performed on sample designs, showing the impact of the PCB stackup (material, drill plan, and so forth) on LPDDR4 performance. The results showed that maximum bandwidth could be achieved on a FR4 solution, but required back-drilling. The higher frequency material could achieve same performance without back drill. Note the 8 layer design only achieved 3733, but this was due to other design compromises due to limited layers (solid reference planes, and so forth).

Table 3-5. LPDDR4 Performance Impact From StackUp

Design	Material	Layer Count	Via Back Drilling	Maximum LPDDR4 Speed (Mbps)
EVM	I-Speed	16	Yes	4266
Ref Board	I-Speed	10	No	4266
Ref Board	370HR	10	Yes	4266
Ref Board	370HR	8	No	3733

3.6.2 Routing

The below examples show the LPDDR4 Clock and CA routing on an example 10 layer PCB design. The clock is routed differentially with target impedance of 70-ohms. For the T-branch to match the impedance of the trace, the impedance needs to be doubled. This can create challenges, as the higher impedances can be difficult to achieve in some PCB stackups. The CA signals are routed targeting 35-ohms, with the T-branch at two times the source impedance.

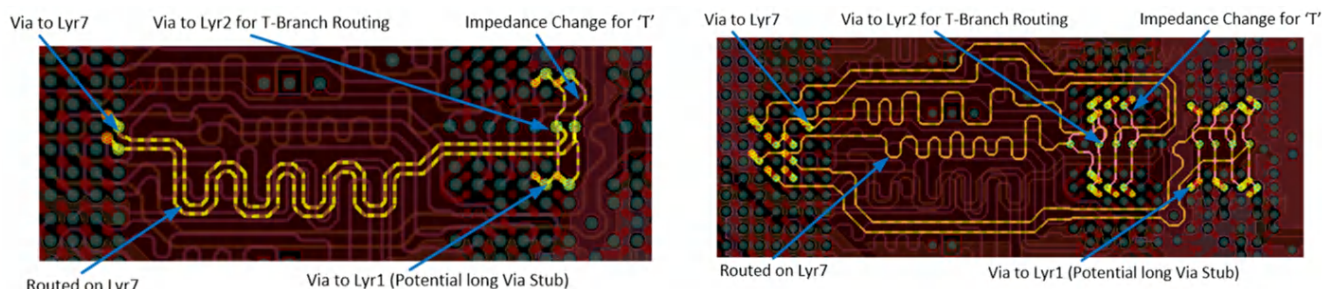


Figure 3-8. Example LPDDR4 Clock and CA Routing

On the same 10 layer reference design, the data groups are routed on layers 2 and 4. The upper layers are used due to the minimum via travel, which minimized the via inductance and via-to-via coupling. Because the data signals are point-to-point, T-branch routing is not required.

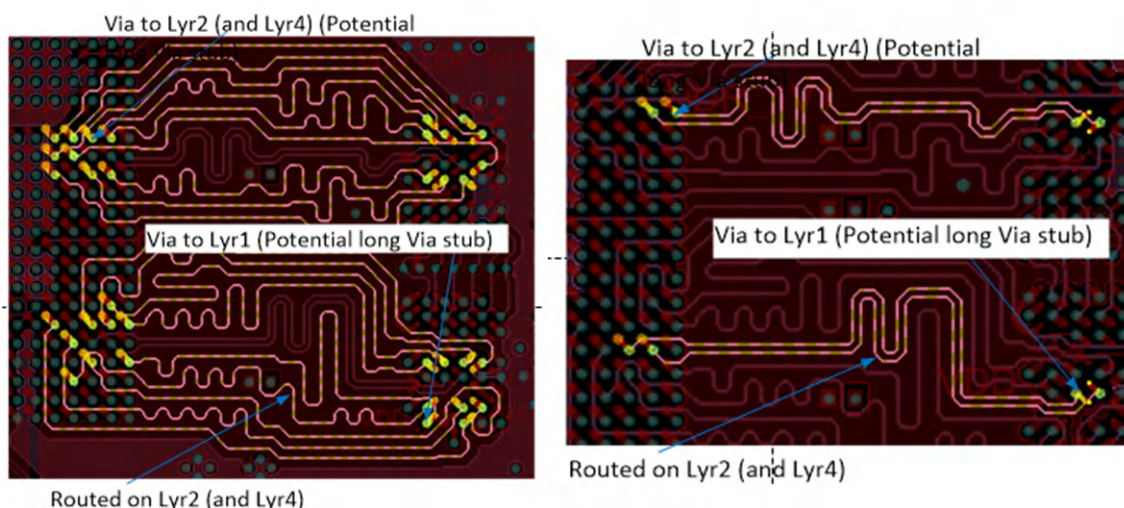


Figure 3-9. Example LPDDR4 Data Byte(s) and DQS(s) Routing

Table 3-6. LPDDR4-4266 Layer Impact Performance (Read at Pad)

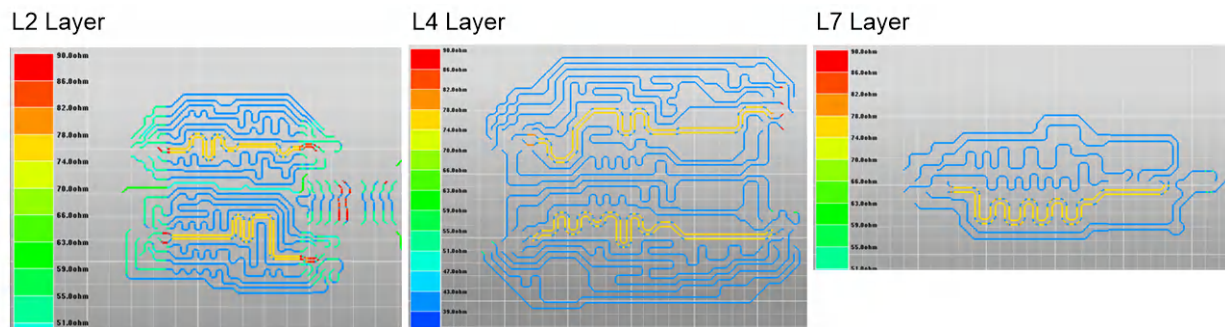
Routing Layer	Via Type	Back Drilling	EW Margin (ps)	EH Margin (mV)
L1, L12	PTH	No	-7.10	56.72
L1, L12	PTH	Yes	-4.86	55.71
L1, L3	PTH	No	5.70	40.29
L1, L3	PTH	Yes	8.37	34.54

Table 3-7. LPDDR4-4266 Layer Impact Performance (Write at Pad)

Routing Layer	Via Type	Back Drilling	EW Margin (ps)	EH Margin (mV)
L1, L12	PTH	No	17.42	39.22
L1, L12	PTH	Yes	20.04	41.93
L1, L3	PTH	No	27.66	41.37
L1, L3	PTH	Yes	27.76	48.63

3.6.3 Model Verification

Before simulating, it is recommended to verify the models. One verification method described is the impedance plot (or impedance scan). The Impedance scans for a 10 layer design are provided.


Figure 3-10. Example LPDDR4 Trace Impedance Scan
Table 3-8. Example LPDDR4 Trace Impedance Summary for Data

Layer	DDR Bus	DQ SE Impedance (Ω)	DQS/CLK Difference Impedance (Ω)
L2	B1 and B3	40.9	77.7
L2	CA	51.7	101.4
L4	B0 and B2	41.1	77.7
L7	CA	41.1	77.7

For CK and CA signals, the goal is to have the branch segment equal to two times the impedance of the feed trace. Note its common for the PCB to limit the achievable impedances. Simulations will show you if the compromises are acceptable.

Table 3-9. Example LPDDR4 Trace Impedance Summary for CA

Board	CA Feed Impedance (Ω)	CA Branch Impedance (Ω)	CA Branch Target (Ω)	Impedance Mismatch (Ω)
Initial Design	49.1	59.6	98 (49x2)	38.6
Final Design	41.1	51.7	82 (41x2)	30.5

The simulation results show the improvement made by closer matching the impedances to their targets.

Table 3-10. Example LPDDR4 Simulation Results From Improved Trace Impedance

Board	Total Eye Width Margin (ps)	Total Eye Height Margin (ps)	Min Ringback Margin H (mV)	Min Ringback Margin L (mV)
Initial Design	58.00	14.00	69.59	18.18
Final Design	124.68	48.08	89.43	25.49

3.6.4 Simulation Results

The simulation results are provided for the LPDDR4 interface from a 10 layer design. These simulation targets must be met to ensure the design will operate at the desired level of performance.

CA simulations need to be verified at the DRAM pin/BGA. This includes:

- Minimum ring-back margins at high/low levels (JEDEC)
- Vix_CK ratio (JEDEC)
- Jitter/noise margins with respect to the eye mask (JEDEC)
- Peak-peak power noise

At DRAM Pin:

Package	Board	Total EW Margin (ps)	Total EH Margin (mV)	Min Rback Margin H (mV)	Min Rback Margin L (mV)	Vix_CK_Ratio (%)
J7 SCK 1207	J7 370HR 10L Ref 121819 ACC, No BD, SIwave	182.04	68.50	89.43	25.50	16.81

Jitter/Noise Eye Margins (points to Total EW Margin and Total EH Margin)
Vix_CA Ratio (points to Vix_CK_Ratio)
Ring-back High/Low Margins (points to Min Rback Margin H and Min Rback Margin L)

Figure 3-11. LPDDR4 Simulation Results for CA

Data write simulations need to be verified at both the DRAM BGA pin and the DRAM pad. This includes:

- Minimum ring-back margins at high/low levels (JEDEC)
- Vix_CK ration (JEDEC)
- Jitter/noise margins with respect to the eye mask (JEDEC)
- Peak-peak power noise

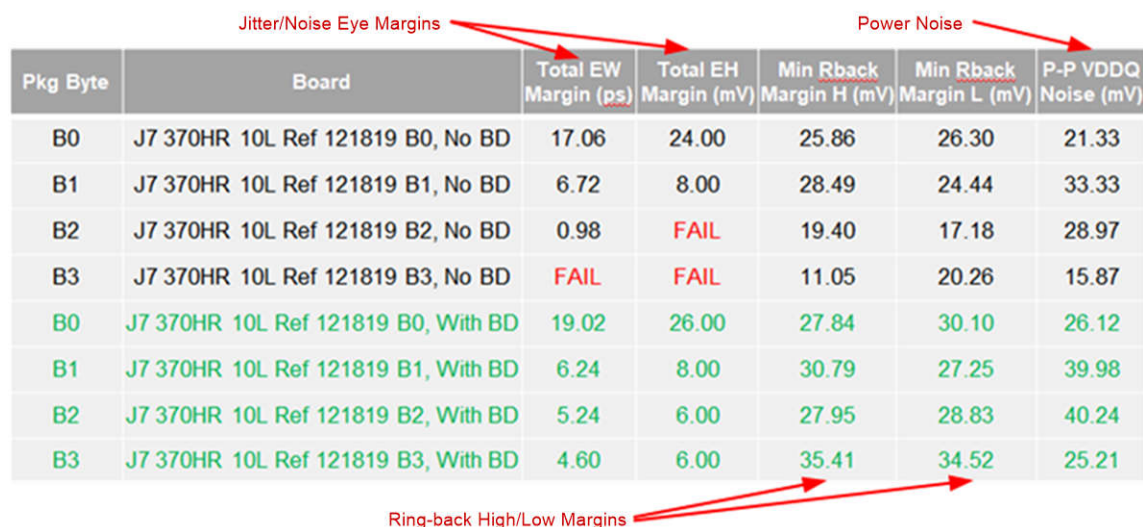
Pkg Byte	Board	Total EW Margin (ps)	Total EH Margin (mV)	Min Rback margin H (mV)	Min Rback Margin L (mV)	Vix_DQS_Ratio (%)
B0	J7 370HR 10L Ref 121819 B0, With BD	62.32	55.20	68.63	111.83	9.80
B1	J7 370HR 10L Ref 121819 B1, With BD	54.52	94.28	86.46	114.69	7.64
B2	J7 370HR 10L Ref 121819 B2, With BD	53.40	73.96	81.19	106.32	5.55
B3	J7 370HR 10L Ref 121819 B3, With BD	54.86	52.74	34.97	48.55	9.81

Jitter/Noise Eye Margins (points to Total EW Margin and Total EH Margin)
Vix_DQS Ratio (points to Vix_DQS_Ratio)
Ring-back High/Low Margins (points to Min Rback margin H and Min Rback Margin L)

Figure 3-12. LPDDR4 Simulation Results for Write

Data read simulations need to be verified at SOC. This includes:

- Minimum ring-back margins at high/low levels
- Jitter/noise margins with respect to the eye mask
- Peak-peak power noise



Pkg Byte	Board	Total EW Margin (ps)	Total EH Margin (mV)	Min Rback Margin H (mV)	Min Rback Margin L (mV)	P-P VDDQ Noise (mV)
B0	J7 370HR 10L Ref 121819 B0, No BD	17.06	24.00	25.86	26.30	21.33
B1	J7 370HR 10L Ref 121819 B1, No BD	6.72	8.00	28.49	24.44	33.33
B2	J7 370HR 10L Ref 121819 B2, No BD	0.98	FAIL	19.40	17.18	28.97
B3	J7 370HR 10L Ref 121819 B3, No BD	FAIL	FAIL	11.05	20.26	15.87
B0	J7 370HR 10L Ref 121819 B0, With BD	19.02	26.00	27.84	30.10	26.12
B1	J7 370HR 10L Ref 121819 B1, With BD	6.24	8.00	30.79	27.25	39.98
B2	J7 370HR 10L Ref 121819 B2, With BD	5.24	6.00	27.95	28.83	40.24
B3	J7 370HR 10L Ref 121819 B3, With BD	4.60	6.00	35.41	34.52	25.21

Figure 3-13. LPDDR4 Simulation Results for Read

The simulations results for read includes two sets for data, black and green. The black shows the design failed, as several bytes failed to meet the eye margins. The green is the simulation results of the same design, but with back-drilling the via stubs applied.

4 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision E (May 2023) to Revision F (August 2024)	Page
• Updated latest routing information in Table 2-6	16
• Updated latest routing information in Table 2-7	18

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